

The primary concept of the LSA is built around a defined set of standards found in 14 CFR part 1, section 1.1:

- Powered (if powered) by single reciprocating engine.
- Fixed landing gear (except seaplanes and gliders).
- Fixed pitch or ground adjustable propeller.
- Maximum takeoff weight of 1,320 pounds for landplane, 1,430 for seaplane.
- Maximum of two occupants.
- Non-pressurized cabin.
- Maximum speed in level flight at maximum continuous power of 120 knots calibrated airspeed (CAS).
- Maximum stall speed of 45 knots. [Figure 17-2]

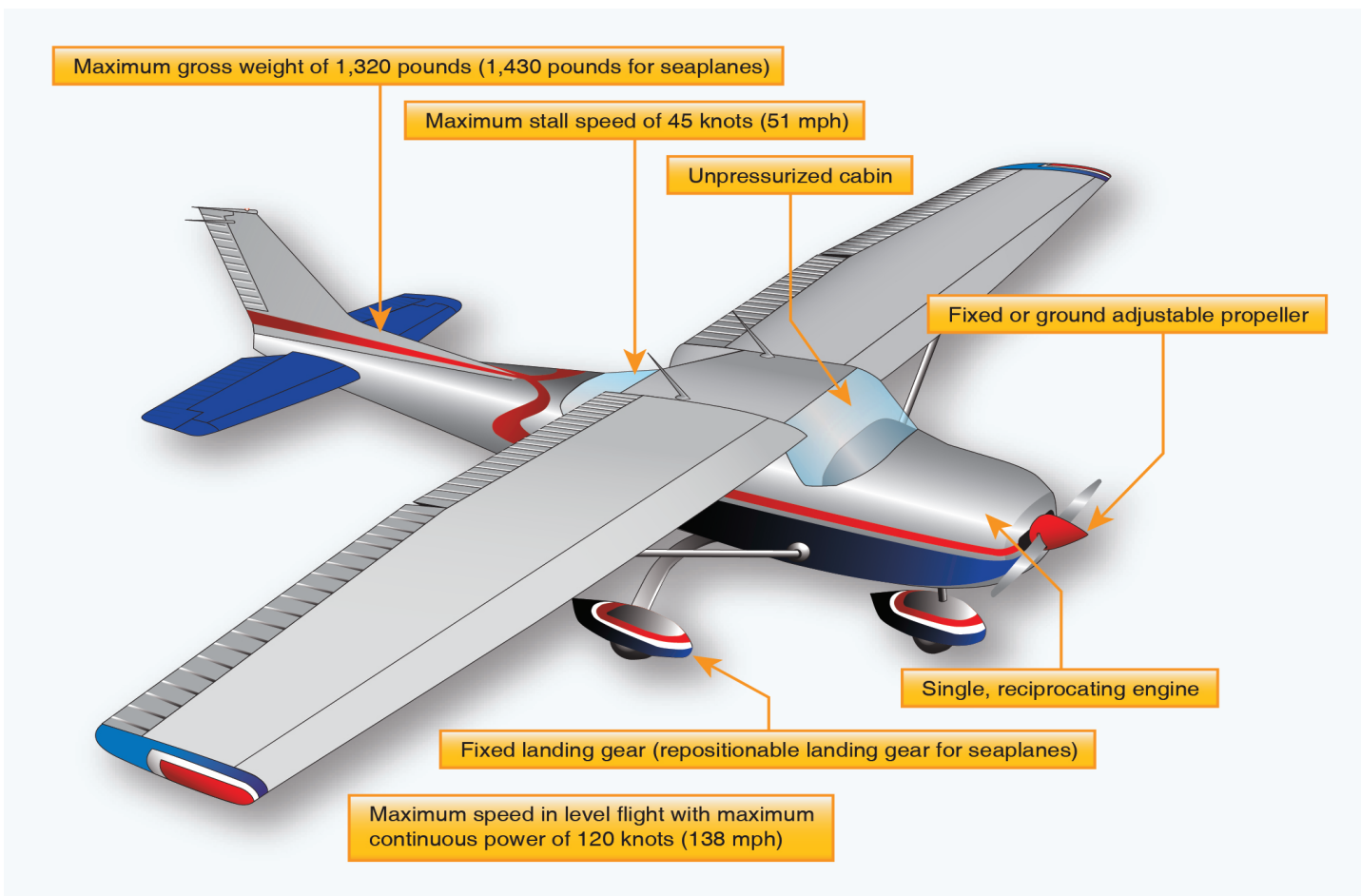


Figure 17-2. *Light-sport airplane.*

The LSA category includes standard, special, and experimental designations. Some standard airworthiness certificated aircraft (i.e., a Piper J-2 or J-3) may meet the Title 14 of the Code of Federal Regulation (14 CFR) part 1, section 1 definition of LSA. Type certificated aircraft that continue to meet the section 1.1 definition of LSA may be flown by a pilot who holds a sport pilot certificate with the appropriate endorsement for the aircraft (14 CFR part 61, section 61.315(a)). The sport pilot certificate is discussed later in this chapter. Aircraft that are specifically manufactured for the LSA market are included in either the Special (S-LSA) or Experimental (E-LSA) designations. An approved S-LSA is manufactured in a ready-to-fly condition and an E-LSA is either a kit or plans-built aircraft based on an approved S-LSA model.

It is important to note that S-LSAs or E-LSAs are not type certificated by the FAA and are not required to meet any airworthiness requirements of 14 CFR part 23. Instead, S-LSAs and E-LSAs are designed and manufactured in accordance with ASTM Committee F-37 Industry Consensus Standards. Therefore, LSA designs are not subjected to the scrutiny, demands, and testing of FAA standard airworthiness certification. Industry Consensus Standards are intended to be less costly and less restrictive than 14 CFR part 23 certification requirements and, as a result, LSA manufacturers have greater latitude with their designs. ASTM Industry Consensus Standards were accepted by the FAA in 2005, which established FAA-accepted industry-developed standards for the design and manufacture of aircraft for the first time.

ASTM Industry Consensus Standards for LSA cover the following areas:

- Design and performance
- Required equipment
- Quality assurance
- Production acceptance tests
- Aircraft operating instructions
- Maintenance and inspection procedures
- Identification and recording of major repairs and major alterations
- Continued airworthiness
- Manufacturer's assembly instructions (E-LSA aircraft)

Using the ASTM Industry Consensus Standards, an LSA manufacturer can design and manufacture their aircraft and assess its compliance to the consensus standards. The manufacturer then, through evaluation services offered by a designated airworthiness representative, completes the process by submitting the required paperwork to the FAA. Upon approval, an LSA manufacturer is permitted to sell ready-to-fly S-LSA aircraft.

Light-Sport Airplane Synopsis

- The airplane must meet the weight, speed, and other criteria listed in 14 CFR part 1, section 1 that define an LSA.
- Airplanes under the S-LSA certification commonly find use in sport and recreation, flight training, and aircraft rental.
- E-LSA-certified airplanes may be used for sport and recreation and flight instruction for the owner of the airplane. However, E-LSA certification is not the same as Experimental Amateur-Built aircraft certification. E-LSA certification is based on an approved S-LSA airplane.
- FAA policy allows sport pilots with an airplane rating to fly certain airplanes (i.e., a Piper J-2 or J-3) that continue to meet the 14 CFR part 1, section 1 LSA definition even though the airplane was originally issued a standard airworthiness type certificate.
- No person may operate the aircraft unless it has been registered by its owner, if eligible for registration, per 14 CFR part 47, section 47.3(b).
- United States or foreign manufacturers can be authorized.
- FAA policy allows holders of a sport pilot certificate or higher level pilot certificate (recreational, private, commercial, or ATP) to pilot sport aircraft.
- LSAs may be operated by VFR at night if the aircraft is equipped with the instruments and equipment specified in 14 CFR part 91, section 91.205(c), and if night operations are allowed by the airplane's operating limitations. However, sport pilots may not fly at night (14 CFR 61.315(c)(5)).
- LSAs may be operated between sunset and sunrise by a recreational pilot as the sole occupant of the aircraft, in accordance with 14 CFR part 61, section 61.101(i)(3) for the purpose of obtaining additional certificates or ratings and while under the supervision of an authorized instructor and provided the flight or surface visibility is at least 5 statute miles.

Sport Pilot Certificate

In addition to the LSA rules, the FAA created a new sport pilot certificate in 2004 that lowered the minimum training time requirements, in comparison to other pilot certificates, for newly certificated pilots wishing to exercise privileges only in LSA aircraft. Pilots who hold recreational, private, commercial, or airline transport pilot certificates may pilot light-sport airplanes provided they possess the appropriate category/class rating and a U.S. driver's license or medical certificate that meets the requirements for the aircraft displayed in the 14 CFR part 61, section 61.303 table. For example, a commercial pilot rated in airplane multiengine land and rotorcraft gyroplane is qualified to fly a light-sport gyroplane as pilot in command (PIC) if also holding a medical certificate or a U.S. driver's license. However, that pilot is not qualified to act as PIC of a light-sport airplane (sport airplanes are single-engine) without supervision from an authorized instructor.

Pilots holding higher level certificates with the appropriate category and class ratings may fly LSAs as long as the pilot holds a valid U.S. driver's license as evidence of medical eligibility. However, if the pilot's most recent medical certificate was denied, revoked, suspended, or withdrawn, a U.S. driver's license is not sufficient. The pilot would then need to hold a valid FAA medical certificate to fly an LSA.

Transition Training Considerations

Flight Schools

The LSA category has created new business opportunities due to low fuel usage, reliability, and low maintenance costs. Many owners and operators of flight schools use S-LSAs for flight instruction and rental.

When considering a transition to LSA, a pilot should look for a flight school that has experience in LSA instruction and can provide quality instruction. Personally touring a school and soliciting feedback from other pilots that have transitioned into LSAs may help find an appropriate school. Some questions to be asked include the following:

- How many pilots has the flight school transitioned into LSAs and how many LSAs are available for instruction?
- What are the flight school's rental, insurance, and safety policies?
- How is maintenance accomplished and by whom?
- How are records maintained and how is scheduling accomplished?

Flight Instructors

The flight instructor is an important link in a successful LSA transition. A transitioning pilot should choose a flight instructor that has verifiable experience in LSA instruction. The Sport Pilot rule allows for a Sport Pilot flight instructor certificate, the flight instructor-S. 14 CFR part 61, section 61.413 limits a flight instructor-S to instruction in LSAs—a flight instructor-S cannot give instruction in a non-LSA airplane (e.g., a Cessna 150). While FAA policy allows a flight instructor certificated as a flight instructor-A to give instruction in both a light-sport airplane and a non-light-sport airplane, a flight instructor-S with teaching experience in LSA might provide better instruction than a flight instructor-A who has minimal teaching experience in light-sport airplanes.

A transitioning LSA pilot should have an opportunity to review the curriculum, syllabus, lesson plans, as well as the process for tracking progress through the training program. Depending on the transitioning pilot's experience, currency, and type of airplane typically flown, the flight instructor should make appropriate adjustments to any LSA training curriculum. A suggested LSA transition training outline is presented:

- CFR review as pertaining to LSAs and sport pilots
- Pilot's Operating Handbook (POH) review
- LSA maintenance
- LSA weather considerations
- Wake turbulence avoidance
- Performance and limitations
- Operation of systems
- Ground operations
- Preflight inspection
- Before takeoff check
- Normal and crosswind takeoff and climb
- Normal and crosswind approach and landing
- Soft-field takeoff, climb, approach, and landing
- Short-field takeoff, climb, approach, and landing
- Go-around/rejected landing
- Steep turns
- Stalls and spin awareness
- Emergency approach and landing
- Systems and equipment malfunctions
- After landing, parking, and securing

LSA Maintenance

LSAs should be treated with the same level of care as any standard airworthiness certificated airplane. However, S-LSAs have greater latitude pertaining to who may conduct maintenance as compared to standard airworthiness certificated airplanes. S-LSAs may be maintained and inspected by:

1. An LSA Repairman with a Maintenance rating; or,
2. An FAA-certificated Airframe and Powerplant Mechanic (A&P); or,
3. As specified by the aircraft manufacturer; or
4. As permitted, owners performing limited maintenance on their S-LSA.

The airplane maintenance manual includes the specific information for repair and maintenance on inspections, repair, and authorization for repairs and maintenance. Most often, S-LSA inspections can be signed off by an FAA-certificated A&P or LSA repairman with a Maintenance rating rather than an A&P with Inspection Authorization (IA); however, the aircraft maintenance manual provides the procedures to follow. The FAA does not issue Airworthiness Directives (ADs) for S-LSAs or E-LSAs. If an FAA-certified component is installed on an LSA, the FAA issues any pertaining ADs for that specific component. Manufacturer safety directives are not distributed by the FAA. S-LSA owners should comply with:

- Safety directives (alerts, bulletins, and notifications) issued by the LSA manufacturer
- ADs if any FAA-certificated components are installed
- Safety alerts (immediate action)
- Service bulletins (recommending future action)
- Safety notifications (informational)

S-LSA compliance with maintenance requirements provides greater latitude for owners and operators of these airplanes. Because of the options in complying with the maintenance requirements, pilots who are transitioning to LSAs should understand how maintenance is accomplished; who is providing the maintenance services; and verify that all compliance requirements have been met.

Airframe and Systems Construction

LSAs may be constructed using wood, tube and fabric, metal, composite, or any combination of materials. In general, the manufacturer selects materials and design to keep the airplane lightweight while maintaining the structural requirements. Composite LSAs tend to be sleek and modern looking with clean lines as molding of the various components allows designers great flexibility in shaping the airframe. Other LSAs are authentic-looking renditions of early aviation airplanes with fabric covering a framework of steel tubes. Of course, LSAs may be anything in between using both metal and composite construction. *[Figure 17-3]* A pilot transitioning into LSA should understand the types of construction and the typical concerns for each type of construction:

- Steel tube and fabric—while the techniques of steel tube and fabric construction hails back to the early days of aviation, this construction method has proven to be lightweight, strong, and inexpensive to build and maintain. Advances in fabric technology continue to make this method of covering airframes an excellent choice. Fabric can be limited in its life span if not properly maintained. Fabric should be free from tears, well-painted with little to no fading, and should easily spring back when lightly pressed.
- Aluminum—an aluminum-fabricated airplane has been a favorite choice for decades. Pilots should be quite familiar with this type of construction. Generally, airframes tend to be lightly rounded structures dotted with rivets and fasteners. This construction is easily inspected due to the wide-spread experience with aluminum structures. Any corrosion, working rivets, dents, and cracks should be identified during a pilot's preflight inspection.
- Composite—a composite airplane is principally made from structural epoxies and cloth-like fabrics, such as bi-directional and uni-directional fiberglass cloths, and specialty cloths like carbon fiber. Airframe components, such as wing and fuselage halves, are made in molds that result in a sculpted, mirror-like finish. Generally, composite construction has few fasteners, such as protruding rivets and bolts. Pilots should become acquainted with inspection concerns such as looking for hair-line cracks and delamination.



Figure 17-3. LSA can be constructed using both metal and composites.

Engines

LSAs use a variety of engines that range from FAA-certificated to non-FAA-certificated. Engine technology varies significantly from conventional air-cooled to high revolutions per minute (rpm)/water-cooled designs. [Figure 17-4] These different technologies present a transitioning pilot new training opportunities and challenges. Since most light-sport airplanes use non-FAA-certificated engines, a transitioning pilot should fully understand the engine controls, procedures, and limitations. In most light-sport airplanes, engines are water-cooled, 4-cycle, and carbureted. These engines have much higher operating rpm and require a gear-box to reduce propeller rpm to the proper range. Because of the higher engine operating rpm, vibration and noise signatures are quite different in most light-sport airplanes when compared to most standard type certificated designs.



Figure 17-4. A water-cooled 4-cycle engine.